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vector

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component form

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transformation

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preimage / image

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translation

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rigid motion

4.2

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reflection

4.2

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line of reflection

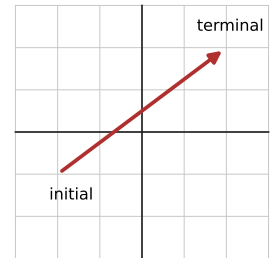
component form

$\langle a, b \rangle$ — the horizontal change a and the vertical change b from the initial point to the terminal point.

From $(-2, -1)$ to $(2, 2)$: $\langle 4, 3 \rangle$

vector

A quantity with both direction and magnitude, drawn from an initial point to a terminal point.



preimage / image

The ORIGINAL figure (input) / the NEW figure after the transformation (output).

$A \rightarrow A'$ (read "A prime")

transformation

A function that moves or changes a figure to produce a new figure.

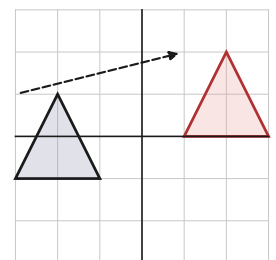
rigid motion

A transformation that preserves length and angle measure. The image is congruent to the preimage.

Translations, reflections, rotations

translation

A transformation that moves every point the same distance in the same direction.



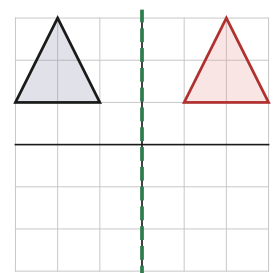
$(x, y) \rightarrow (x + a, y + b)$

line of reflection

The mirror line. It is the perpendicular bisector of every segment joining a point to its image.

reflection

A transformation that uses a line like a mirror to flip a figure.



In the y -axis: $(a, b) \rightarrow (-a, b)$

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glide reflection

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line symmetry

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line of symmetry

4.3

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rotation

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center of rotation

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angle of rotation

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rotational symmetry

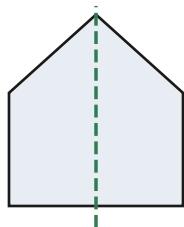
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center of symmetry

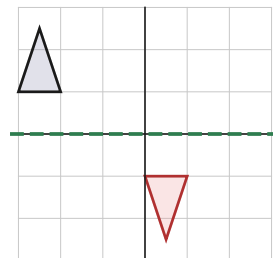
line symmetry

A figure has line symmetry when a reflection in some line maps the figure onto itself.



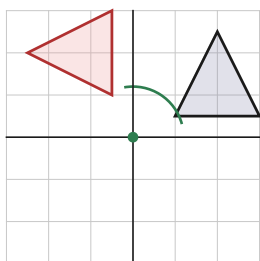
glide reflection

A translation followed by a reflection in a line PARALLEL to the translation vector.



rotation

A transformation in which a figure is turned about a fixed point.



90° about O : $(a, b) \rightarrow (-b, a)$

line of symmetry

The line of reflection that maps a figure onto itself. A figure may have more than one.

Rectangle (not a square): 2. Square: 4.

angle of rotation

The angle formed by rays drawn from the center to a point and to its image.

Counterclockwise unless told otherwise.

center of rotation

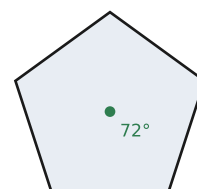
The fixed point the figure turns about.

center of symmetry

The center point of a rotation that maps a figure onto itself (see rotational symmetry).

rotational symmetry

A figure has it when a rotation of 180° or less about its center maps the figure onto itself.



Regular pentagon: every 72°

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**composition of
transformations**

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congruent figures

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**congruence
transformation**

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dilation

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center of dilation

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scale factor

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similar figures

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**similarity
transformation**

congruent figures

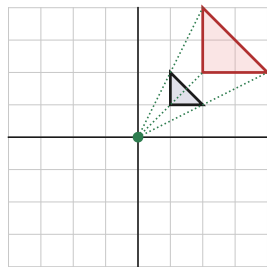
Figures with the same size and shape.

composition of transformations

Two or more transformations combined into a single transformation. Do them in order; label A' , then A'' .

dilation

A transformation that enlarges or reduces a figure by a scale factor, from a fixed center.



$$(x, y) \rightarrow (kx, ky)$$

congruence transformation

A transformation (or composition of rigid motions) that maps a figure onto a congruent figure.

scale factor

The ratio k of an image length to the corresponding preimage length.

$k > 1$: enlargement. $0 < k < 1$: reduction.

center of dilation

The fixed point a dilation stretches or shrinks from. Every image point lies on a ray from the center.

similarity transformation

A dilation, or a dilation combined with rigid motions. It maps a figure onto a similar figure.

similar figures

Figures with the same shape but not necessarily the same size.

Corresponding angles $\cong$; sides proportional